Arunabha Majumdar

Curriculum Vitae

Office MA207, Math-Physics-LA building
IIT Hyderabad, Kandi, Telangana, India
DOB: 11th Oct, 1984 ☎ (+91)7075484688
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Work Experience

January, 2021–present **Assistant Professor**.
Department of Mathematics, Indian Institute of Technology (IIT) Hyderabad, India

2019–2020 **Assistant project scientist**.
Department of Pathology and Laboratory Medicine
David Geffen School of Medicine
University of California, Los Angeles, USA

2018–2019 **Postdoctoral scholar**.
Department of Pathology and Laboratory Medicine
David Geffen School of Medicine
University of California, Los Angeles, USA
Mentor: Prof. Bogdan Pasaniuc

2014–2017 **Postdoctoral scholar**.
Department of Epidemiology and Biostatistics
University of California, San Francisco, USA
Mentor: Prof. John S. Witte

Education

2007–2013 **Ph.D. in Statistics - Indian Statistical Institute Kolkata**, India.
Thesis Title: Some statistical issues pertaining to genome-wide association studies
Supervisor: Late Prof. Saurabh Ghosh

2005–2007 **M.Sc. in Statistics, Indian Institute of Technology Kanpur**, India.
Grade: Good (8.4/10)

2002–2005 **B.Sc. in Statistics, Ramakrishna Mission Residential College Narendrapur, Calcutta University**, West Bengal.
Grade: Excellent (75% in Statistics honors), ranked fourth in the Calcutta University

Honors and Awards

- Awarded the SERB international travel support (ITS) to present research in the Royal Statistical Society (RSS) International Conference, 2023, Harrogate, UK.
- International Genetic Epidemiology Society (IGES) 2012 travel award to attend IGES 2012 meeting in Portland, USA.
- Genetic Analysis Workshop (GAW) 18 travel award to attend GAW 18 workshop, in Portland,

USA.

- Nominated for the Neel award for best postdoc oral presentation in the International Genetic Epidemiology Society (IGES) meeting in San Diego, USA (October 2018).
- Nominated for the Neel award for best postdoc oral presentation in the International Genetic Epidemiology Society (IGES) meeting in Toronto, Canada (October 2016).

Research Interests

- Statistical genomics and biostatistics
- Statistical challenges in transcriptome-wide association study (TWAS)
- Statistical methods for gene-environment interaction study
- Methods for sub-typing complex diseases based on genetics and environment
- Methods for evaluating replicability of discoveries
- Fine mapping in TWAS

Publications

$^{\$}$corresponding author, *co-first author.

20. S Mishra, A Majumdar$^{\$}$. A Multi-phenotype approach to joint testing of main genetic and gene-environment interaction effects. *Statistics in Medicine*, 2025. Vol. 44 (20-22), e70253, DOI: https://doi.org/10.1002/sim.70253

19. K Kundal, KV Rao, A Majumdar, N Kumar, R Kumar. Comprehensive benchmarking of CNN-based tumor segmentation methods using multimodal MRI data. *Computers in Biology and Medicine*, 2024. Vol. 178, DOI:10.1016/j.compbiomed.2024.108799.

18. A Majumdar$^{\$}$, B Pasaniuc. A Bayesian method for estimating gene-level polygenicity under the framework of transcriptome-wide association study. *Statistics in Medicine*, 2023. Vol. 42(26), 4867-4885, DOI:10.1002/sim.9892.

17. S Lindstroem*, L Wang*, H Feng*, A Majumdar*, S Guo*, J Macdonald et al. Genome-wide analyses characterize shared heritability among cancers and identify novel cancer susceptibility regions. *Journal of the National Cancer Institute*, 2023. Vol. 115(6), 712-732, DOI:10.1093/jnci.

16. J Das, BH Beyaztas, MK Mac-Ocloo, A Majumdar, A Mandal. Testing equality of multiple population means under contaminated normal model using the density power divergence. *Entropy*, 2022. Vol. 24(9), 1189, DOI:10.3390/e24091189.

15. A Majumdar*, P Patel*, B Pasaniuc, RA Ophoff. A summary-statistics based approach to examine the role of serotonin transporter promoter tandem repeat polymorphism in psychiatric phenotypes. *European Journal of Human Genetics*, 2022. Vol. 30(5), 547-554, DOI:10.1038/s41431-021-00996-6.

14. A Majumdar, S Ghosh. Competing analytical strategies for combining associated single nucleotide polymorphisms for estimating genetic risks. *Journal of Genetics*, 2021. Vol. 101(1), 14.

13. H Chen, A Majumdar, L Wang, . . . , P Kraft, B Pasaniuc, S Lindstrom. Large-scale cross-cancer fine-mapping of the 5p15.33 region reveals multiple independent signals. *Human Genetics and Genomics Advances*, 2021. Vol. 2(3), 100041, DOI:10.1016/j.xhgg.2021.100041.

12. A Majumdar[$], C Giambartolomei, N Cai, T Haldar, T Schwarz, M Gandal, J Flint, B Pasaniuc[$]. Leveraging eQTLs to identify individual-level tissue of interest for a complex trait. *PLOS Computational Biology*, 2021. Vol. 17(5), e1008915, DOI:10.1371/journal.pcbi.1008915.

11. H Feng, N Mancuso, A Gusev, A Majumdar, M Major, B Pasaniuc, P Kraft. Leveraging expression from multiple tissues using sparse canonical correlation analysis (sCCA) and aggregate tests improve the power of transcriptome-wide association studies (TWAS). *PLOS Genetics*, 2021. Vol. 17(4), e1008973. DOI:10.1371/journal.pgen.1008973.

10. A Majumdar, KS Burch, S Sankararaman, B Pasaniuc, W Gauderman, JS Witte. A two-step approach to testing overall effect of gene-environment interaction for multiple phenotypes. *Bioinformatics*, 2021. Vol. 36(24), 5640-5648, DOI:10.1093/bioinformatics/btaa1083.

9. I Mandric, T Schwarz, A Majumdar, K Hou, L Briscoe, R Perez, M Subramaniam, C Hafemeister, R Satija, C Ye, B Pasaniuc, E Halperin. Optimal design of single-cell RNA sequencing experiments for cell-type-specific eQTL analysis. *Nature Communications*, 2020. Vol. 11(1), 5504, DOI:10.1038/s41467-020-19365-w.

8. K Hou*, KS Burch*, A Majumdar, H Shi, N Mancuso, Y Wu, S Sankararaman, B Pasaniuc. Accurate estimation of SNP-heritability from biobank-scale data irrespective of genetic architecture. *Nature Genetics*, 2019. Vol. 51(8), 1244-1251. DOI:10.1038/s41588-019-0465-0.

7. A Majumdar, T Haldar, S Bhattacharya, JS Witte. An efficient Bayesian meta-analysis approach for studying cross-phenotype genetic associations. *PLOS Genetics*, 2018, Vol. 14(2), 1-32. DOI:10.1371/journal.pgen.1007139.

6. JD Hoffman, RE Graff, NC Emami, CG Tai, MN Passarelli, D Hu, S Huntsman, D Hadley, L Leong, A Majumdar, N Zaitlen, E Ziv, JS Witte. Cis-eQTL-based trans-ethnic meta-analysis reveals novel genes associated with breast cancer risk. *PLOS Genetics*, 2017, Vol. 13(3), 1-19. DOI:10.1371/journal.pgen.1006690

5. A Majumdar, T Haldar, JS Witte. Determining which phenotypes underlie a pleiotropic signal. *Genetic Epidemiology*, 2016, Vol. 40(5), 366-381. DOI: 10.1002/gepi.21973.

4. A Majumdar[$], JS Witte, S Ghosh. Semi-parametric allelic tests for mapping multiple phenotypes: Binomial regression and Mahalanobis distance. *Genetic Epidemiology*, 2015, Vol. 39(8), 635-650. DOI: 10.1002/gepi.21930.

3. A Majumdar, I Mukhopadhyay, S Ghosh. Association mapping of blood pressure levels in a longitudinal framework using Binomial regression. *BMC Proceedings*, 2014, Vol. 8(1), 1-4. BioMed Central. DOI: 10.1186/1753-6561-8-S1-S74.

2. A Majumdar, S Ghosh. A multivariate normal block versus a principal components approach: competing strategies for multiple testing in a genome-wide case-control association framework. *Journal of the Indian Society of Agricultural Statistics*, 2014, Vol. 68(2), 217-225.

1. A Majumdar, S Bhattacharya, A Basu, S Ghosh. A novel Bayesian semi-parametric algorithm for inferring population structure and adjusting for case control association tests. *Biometrics*, 2013, Vol. 69(1), 164-173. DOI: 10.1111/biom.12004.

Software packages developed

- A Majumdar, T Haldar. R package MPGE: A Two-Step Approach to Testing Overall Effect of Gene-Environment Interaction for Multiple Phenotypes, 2020. `https://cran.r-project.org/web/packages/MPGE/index.html`
- A Majumdar, T Haldar, B Pasaniuc. R package eGST: Leveraging eQTLs to Identify Individual-Level Tissue of Interest for a Complex Trait, 2019. `https://cran.r-project.org/web/packages/eGST/index.html`
- A Majumdar, T Haldar, JS Witte. R package CPBayes: Bayesian Meta Analysis for Studying Cross-Phenotype Genetic Associations, 2018. `https://CRAN.R-project.org/package=CPBayes`.

Teaching experience

I have taught various statistics and mathematics courses in the undergraduate, postgraduate, and Ph.D. programs at Indian Institute of Technology (IIT) Hyderabad.

- Monsoon, 2025: Introduction to Bayesian Statistics
- Winter, 2025: Topics in Statistics (1 credit as a part of the Ph.D. course).
- Winter, 2025: Applied Statistics
- Monsoon, 2024: Bayesian theory and computation
- Winter 2024: Introduction to Bayesian Statistics
- Winter 2024: Applied Statistics (1 credit, taught one-third of the course)
- Monsoon, 2023: Introduction to Time Series Analysis (2 credits, taught two-third of the course)
- Winter, 2023: Introduction to Bayesian Statistics
- Monsoon, 2022: Introduction to Time Series Analysis
- Winter, 2022: Introduction to Statistics (1 credit), 250 students
- Winter, 2022: Introduction to Bayesian Statistics
- Monsoon, 2021: Calculus II (1 credit, 250 students)
- Winter, 2021: Introduction to Statistics (1 credit, 250 students)
- Winter, 2021: Introduction to Bayesian Statistics

Supervision

Ph.D. students
- Saurabh Mishra (5^{th} year ongoing), IIT Hyderabad, MoE fellowship.
- Rehan Khan (3^{rd} year ongoing), IIT Hyderabad, MoE fellowship, jointly with Dr. Sameen Naqvi.
- Ishita Agrawal (2^{nd} year ongoing), IIT Hyderabad, MoE fellowship.

M.Sc. project students
- Rima Rati Patra [completed]
- Yasaswini Nasaka [completed]
- Santosh Kumar [completed]
- Rupa Kumari [completed]

B.Tech. project students
- Ananya Reddy [completed]

Reviewer in journals

- American Journal of Human Genetics
- Annals of Applied Statistics
- Bioinformatics (2)
- BMC Bioinformatics
- Genetic Epidemiology
- European Journal of Human Genetics
- Molecular Neurobiology
- Nature Communications
- Neuropsychopharmacology & Biological Psychiatry
- PLoS Genetics (2)
- Sankhya B
- Scientific Reports
- Statistics and probability letters
- VirusDisease

Funding

- SERB international travel support to present research in the Royal Statistical Society (RSS) International Conference, 2023, Harrogate, UK. Total funding: 1.7 lakhs.
- Seed grant of INR 13 lakhs for two years from IIT Hyderabad. Project title: Bayesian statistical approaches to study gene-level polygenicity and pleiotropy for complex phenotypes.

Administrative responsibilities

- Faculty search committee member: Sept 2025 –
- Departmental Undergraduate Committee (DUGC) member since the academic year 2023 – present.
- Faculty advisor for B. Tech. in Mathematics and Computing 2023 batch – present.
- Timetable committee for Mathematics department: Academic year 2023, 2024, 2025.
- Selection committee member for recruiting office staffs in the Mathematics department: 2023, 2024, 2025.
- Ph.D. admission coordinator: January 2022 semester and July 2022 semester

Serving on doctoral committee

- Arpan Singh (Statistics) [completed]
- Teega Venkateswararao (Statistis) [completed]
- Ravi Kumar (Statistics) [completed]
- Syamantak Das (Mathematics) [completed]
- Kavita Kundal (Biotechnology)
- Bhanu Teja Korra (Biotechnology)
- Swarna Prabha Maharana (Physics)
- Salvadi Chetan Kumar (Civil engineering)
- Daneti Arun Sourya (Civil engineering)

Presentations at conferences

Talk

- Indian Society of Human Genetics meeting 2025, NIMHANS, Bangalore, India. Title of the talk: A unified Bayesian approach to transcriptome-wide association study (TWAS).
- International Indian Statistical Association meeting 2024, Cochin, Kerala. Title of the talk: A Bayesian method for estimating gene-level polygenicity under the framework of transcriptome-wide association study.
- European Mathematical Genetics Meeting (EMGM) 2023, University of Surrey, UK. Title of the online talk: A Bayesian method for estimating gene-level polygenicity under the framework of transcriptome-wide association study.
- International Indian Statistical Association (IISA) meeting 2022, Bengaluru, India. Title: Adjustment for uncertainty of predicted expression in transcriptome-wide association study: a fusion of measurement error theory and bootstrapping.
- Joint Statistical Meetings (JSM) 2019, Denver, USA. Title: Leveraging eQTLs to identify individual-level tissue of interest for a complex trait.
- American Society of Human Genetics (ASHG) Meeting 2018, San Diego, USA. Title: Leveraging eQTLs to identify individual-level tissue of interest for a complex trait.
- International Genetic Epidemiology Society (IGES) Meeting 2018, San Diego, USA. Title: Leveraging eQTLs to identify individual-level tissue of interest for a complex trait.
- International Genetic Epidemiology Society (IGES) Meeting 2016, Toronto, Canada. Title: An efficient Bayesian meta-analysis approach for studying cross-phenotype genetic associations.

Poster

- International Genetic Epidemiology Society (IGES) Virtual Meeting 2021. Title: A summary-statistics based approach to examine the role of serotonin transporter promoter tandem repeat polymorphism in psychiatric phenotypes.
- American Society of Human Genetics (ASHG) Meeting 2016, Vancouver, Canada. Title: An efficient Bayesian meta-analysis approach for studying cross-phenotype genetic associations.
- American Society of Human Genetics (ASHG) Meeting 2015, Baltimore, USA. Title: Determining which phenotypes underlie a pleiotropic signal.
- International Genetic Epidemiology Society (IGES) Meeting 2015, Baltimore, USA. Title: Semi-parametric allelic tests for mapping multiple phenotypes: Binomial regression and Mahalanobis distance.
- International Genetic Epidemiology Society (IGES) Meeting 2012, Stevenson, USA. Title: A novel Bayesian semi-parametric algorithm for inferring population structure and adjusting for case control association tests.
- International Genetic Epidemiology Society (IGES) Meeting 2011, Heidelberg, Germany. Title: A multivariate normal block versus a principal components approach: competing strategies for multiple testing in a genome-wide case-control association framework.

Invited talks

- Invited talk in the symposium to celebrate the foundation day of the National Institute of Biomedical Genomics (2025). Title: A multi-phenotype approach to joint testing of main genetic and gene-environment interaction effects.
- Organized by IISER Thiruvananthapuram (2024): 7-day high-end SERB-Karyashala workshop on 'Real-life Data Modeling via Statistical and Machine Learning Tools'
  - Lecture 1: A brief introduction to genome-wide and transcriptome-wide association studies.
  - Lecture 2: A Bayesian method for estimating gene-level polygenicity under the framework of transcriptome-wide association study.
- Seminar in the Department of Mathematics at Indian Institute of Technology (IIT) Guwahati (2022): Adjustment for the uncertainty of predicted expression in transcriptome-wide association study (TWAS): A fusion of measurement error theory and bootstrapping.
- Seminar in the Department of Statistics at Calcutta University (2023): A Bayesian method for estimating gene-level polygenicity under the framework of transcriptome-wide association study.
- Seminar in the Department of Statistics at the Presidency University (2023): A Bayesian method for estimating gene-level polygenicity under the framework of transcriptome-wide association study.

Workshops attended

- Joint UCLA and Stanford Statistical Genetics Course and Programming Workshop (August, 2016).
- Winter School on a Genetic Dissection of Complex Disease: Analytical Approach organized by National Institute of Biomedical Genomics, Kalyani, West Bengal, India (January, 2012).
- Summer School on Statistical Genomics organized by the institute for mathematical Sciences, National University of Singapore (June, 2009).
- Human Genome Organization Satellite Meeting on Complex Diseases: Approaches to Gene Identification and Therapeutic Management, Kolkata, India (September, 2008).

Outreach activity

I have organized offline seminars in Statistics in the Department of Mathematics, IIT Hyderabad. I have arranged the visits of the following scientists to the campus:

1. Dr. Banoth Veeranna, Ph.D. from the Central University of Hyderabad. Talk title: Bivariate Pseudo Modeling
2. Dr. Palash Ghosh, Assistant professor in the Department of Mathematics, IIT Guwahati.
   - Talk 1: Introduction to Clinical Trials: Statistical Perspectives
   - Talk 2: Towards Personalized Medicine: Sequential Multiple Assignment Randomized Trials (SMARTs)
3. Dr. Priyanka Majumder, Assistant professor in the School of Data Science, IISER Thiruvananthapuram:
   - Talk 1: Sample size requirements to detect an intervention by time interaction in four level longitudinal cluster randomized trials.
   - Talk 2: Detecting trend change in hazard functions – an L-statistic approach

Languages

Bengali	Native speaker
English	Fluent

Hindi	Proficient

Hobbies

- Listening to music
- Watching movies
- Reading novels

References

- Prof. John S. Witte, currently Professor at Stanford University
  Email: jswitte@stanford.edu
- Prof. Bogdan Pasaniuc, currently Professor at University of Pennsylvania
  Email: bogdan.pasaniuc@pennmedicine.upenn.edu